

Temporally Aware Citation Link Prediction in Library and Information Science

Moses Boudourides¹ and Giannis Tsakonas²

¹School of Professional Studies, Northwestern University, Evanston, IL, USA

`Moses.Boudourides@northwestern.edu`

²Library & Information Center, University of Patras, Greece

`gtsak@upatras.gr`

Abstract

Citation link prediction is a central problem in bibliometrics, yet machine-learning approaches in this domain are frequently compromised by temporal leakage, whereby future network topology or retrospectively updated metadata influence predictions of past citation behavior. In this study, we introduce a strongly leakage-mitigated methodological framework for citation link prediction grounded explicitly in the temporal structure of scholarly communication. We formalize four forms of temporal leakage—future citation leakage, future topology leakage, metadata leakage, and semantic leakage—and construct a temporally constrained feature-engineering pipeline in which all network-derived quantities are computed exclusively from information available prior to the publication year of the citing paper. We additionally provide formal definitions of the temporally capped graph G_{t_A-1} , the constrained candidate space $C(A)$, and the ranking objective underlying retrieval-oriented evaluation. The framework is evaluated on two large-scale Library and Information Science (LIS) datasets: Dimensions (662,094 candidate pairs) and OpenAlex (478,698 candidate pairs). These datasets are treated as complementary experiments rather than as a strictly controlled comparison: Dimensions functions as a structural-only baseline, whereas OpenAlex enables a semantic+structural framework through abstract availability. Evaluation is organized across four levels of task complexity: pairwise discriminability, top- k retrieval realism, recommendation realism, and candidate-space complexity. Gradient Boosting achieves ROC-AUC scores of 0.895 on Dimensions and 0.976 on OpenAlex, with performance remaining stable across four rolling temporal evaluation windows. To move beyond binary discrimination, we further implement top- k ranking experiments over temporally and topically constrained candidate pools, obtaining NDCG@10 scores of 0.856 and 0.937 respectively, together with 95% bootstrap confidence intervals for ranking metrics. Feature ablation reveals that while prestige and structural overlap provide meaningful baseline enrichment, semantic similarity overwhelmingly dominates predictive performance once abstract text is available. Under these conditions, citation prediction increasingly behaves less like a purely network-theoretic task and more like a semantically guided retrieval process augmented by structural contextualization. More broadly, the results demonstrate that citation behavior remains highly predictable even under temporally rigorous evaluation, suggesting that scholarly attention is concentrated within relatively stable semantic and structural neighborhoods. The study therefore contributes not only a predictive framework, but also a methodological foundation for leakage-aware evaluation in bibliometric machine learning.

1 Introduction

Scientific citations constitute the primary mechanism through which scholarly knowledge is accumulated, validated, and integrated into the scientific record. Citation networks therefore encode not only the intellectual lineage of individual publications but also the evolving structure of scientific communities and disciplinary attention. Predicting the formation of new links within these networks—citation link prediction—has consequently emerged as a central problem at the intersection of informetrics, network science, information retrieval, and machine learning (Wang et al., 2013; Shibata et al., 2012; He et al., 2010; Beel et al., 2016). Accurate citation prediction has practical applications in literature recommendation systems, research-front detection, and scholarly search, while also contributing to the broader theoretical understanding of how scientific attention is allocated across increasingly large and fragmented bodies of published knowledge.

At the same time, the methodological validity of citation prediction remains deeply sensitive to temporality. Citation networks are cumulative and dynamically evolving systems in which the information available at the time of citation differs fundamentally from the information observable retrospectively. Nevertheless, many machine learning studies construct features from the fully aggregated static graph, thereby allowing future information to influence past predictions implicitly. For example, computing a paper’s PageRank on the complete 2024 citation network and using it to predict a citation that occurred in 2010 allows the model to exploit information about the paper’s eventual scientific success that was unavailable to the citing author at the time of writing. Such future topology leakage artificially inflates predictive performance and weakens causal interpretability by obscuring the historically available information that actually governed citation behavior. Temporal leakage has become an increasingly important concern in applied machine learning more broadly, particularly in temporally ordered prediction tasks (Fortunato et al., 2018), yet citation prediction remains especially vulnerable because the temporal structure of bibliometric data is frequently ignored in benchmark construction and feature engineering.

Accordingly, the present study is positioned not primarily as a new recommendation architecture, but as a strongly leakage-mitigated evaluation framework for bibliometric citation prediction. Our primary contribution is methodological. We introduce a formal taxonomy of temporal leakage and provide explicit mathematical definitions of the temporally capped graph, the constrained candidate space, and the ranking objective. We engineer a feature set encompassing structural overlap, prestige, temporal dynamics, and semantic similarity, while ensuring that all network-derived quantities are computed exclusively from information available prior to the publication year of the citing paper. We pair this feature construction with a temporally constrained evaluation protocol incorporating hard negative sampling, rolling temporal windows, and top- k ranking experiments designed to establish a rigorous intermediate step between pairwise discrimination and large-scale retrieval evaluation.

Our secondary contribution is the application of this framework to the discipline of Library and Information Science (LIS) using two major bibliographic infrastructures: Dimensions.ai and OpenAlex. We treat these datasets as complementary experiments rather than as a perfectly controlled comparison because their metadata exposures differ fundamentally. Dimensions provides a structural-only baseline derived primarily from citation topology and metadata, whereas OpenAlex enables the construction of a semantic+structural framework through abstract avail-

ability. This asymmetry allows us to investigate directly the relative predictive contributions of prestige, structural overlap, temporal information, and semantic similarity under leakage-mitigated conditions.

Our principal empirical finding is that semantic similarity overwhelmingly dominates the predictive signal once abstract text is available. In the OpenAlex framework, the contribution of textual similarity substantially exceeds that of higher-order graph structure, to the point where citation prediction increasingly resembles a semantic retrieval task rather than a purely network-theoretic prediction problem. Rather than treating this outcome as a methodological weakness, we interpret it as an important empirical observation about citation behavior in abstract-rich corpora and about the relative roles of semantics and topology in bibliometric prediction.

Finally, we explicitly delimit the scope of the present study. Our retrieval experiments approximate recommendation realism through constrained candidate pools rather than full-corpus search; our graph-theoretic features function primarily as baseline enrichment rather than as sophisticated higher-order topological descriptors; and our empirical analysis is restricted to LIS. In addition, because TF-IDF representations are fitted on the complete corpus vocabulary, the framework should be understood as strongly leakage-mitigated rather than perfectly leakage-free. These limitations are acknowledged explicitly in order to distinguish clearly between the methodological guarantees established here and broader claims that remain outside the scope of the present work.

2 Literature Review

2.1 Citation Dynamics and Network Evolution

The dynamics of scientific citation have long been associated with cumulative advantage processes in which highly cited papers attract future citations at disproportionate rates. Merton formalized this mechanism sociologically as the Matthew Effect (Merton, 1968), while network science later operationalized it mathematically through preferential attachment models (Barabási and Albert, 1999). In citation networks, preferential attachment implies that the probability of acquiring new citations grows as a function of existing citation counts. However, degree accumulation alone cannot fully explain citation behavior because scientific attention is simultaneously shaped by aging, obsolescence, topical relevance, and disciplinary visibility (Price, 1965). Wang, Song, and Barabási (Wang et al., 2013) demonstrated that long-term citation trajectories emerge from the interaction among preferential attachment, intrinsic fitness, and temporal aging. These mechanisms directly motivate the present framework: historical prestige operationalizes cumulative advantage, semantic similarity approximates topical fitness, and temporal distance captures aging and obsolescence effects.

2.2 Temporal Link Prediction and Leakage

Temporal link prediction—predicting which edges will form in a dynamic network given its history up to time t (Liben-Nowell and Kleinberg, 2007)—is a well-established problem in network science. In citation networks, temporality is not merely methodological but fundamentally causal, since a citation can only occur after the publication of the cited work. Nevertheless, many citation-prediction studies construct features retrospectively from fully aggregated graphs,

thereby allowing future information to influence past predictions implicitly. Such temporal leakage artificially inflates predictive performance and weakens causal interpretability by obscuring the historically available information that governed citation behavior at the time the citation decision was made.

Prior studies have explored structural overlap measures and temporal features in citation prediction (Shibata et al., 2012; Chakraborty et al., 2015), but relatively few have formalized leakage explicitly as a methodological problem. More broadly, leakage-aware evaluation has become an increasingly important concern in applied machine learning because temporally ordered systems are especially vulnerable to hidden future-information contamination (Fortunato et al., 2018). These concerns are particularly acute in bibliometric prediction because static citation graphs implicitly encode future scientific prominence. The present study extends this literature by introducing an explicit taxonomy of temporal leakage together with formally defined temporally capped graph representations and rolling temporal evaluation procedures.

2.3 Bibliographic Coupling and Structural Proximity

Bibliographic coupling (Kessler, 1963) and co-citation analysis (Small, 1973) remain foundational measures of intellectual proximity in bibliometrics. Bibliographic coupling captures shared intellectual ancestry through common references, whereas co-citation reflects the extent to which subsequent literature jointly associates two papers. Both measures have been widely applied to research-front detection, science mapping, and scholarly clustering.

Within citation prediction, these mechanisms provide structural representations of topical and disciplinary proximity embedded within the evolving citation graph. Structural overlap features derived from shared references and shared citers therefore complement semantic textual similarity by capturing relationships perceived through the existing literature itself. In the present framework, bibliographic coupling and co-citation are incorporated explicitly as temporally constrained structural features whose predictive contribution can be isolated through ablation analysis.

2.4 Machine Learning and Semantic Citation Prediction

Early machine learning approaches to citation prediction relied primarily on local structural features combined with linear classifiers (Liben-Nowell and Kleinberg, 2007). More recent work increasingly incorporates semantic representations derived from titles, abstracts, and full text. Traditional vector-space approaches based on TF-IDF representations remain widely used because of their interpretability and computational transparency, while more recent systems employ contextual neural embeddings such as BERT and SciBERT (Devlin et al., 2019).

Content-based citation recommendation has demonstrated that semantic information substantially improves predictive performance beyond topology alone (Bhagavatula et al., 2018). Ensemble learning methods, including Random Forest (Breiman, 2001) and Gradient Boosting (Natekin and Knoll, 2013), have similarly shown strong predictive performance in bibliometric tasks characterized by complex non-linear interactions among prestige, structure, semantics, and temporality. Kozłowski et al. (Kozłowski et al., 2025) further demonstrated that semantic similarity is among the strongest predictors of citation behavior, rivaled primarily by direct social or collaborative ties. The present study builds upon this literature while evaluating semantic and structural signals under explicitly leakage-mitigated temporal constraints.

2.5 Information Retrieval Evaluation and Learning-to-Rank

Binary classification metrics such as ROC-AUC and F1-score provide only partial evaluation for citation recommendation systems because scholarly recommendation is fundamentally a ranking problem (Manning et al., 2008; Powers, 2011). Information retrieval frameworks developed in large-scale evaluation settings such as TREC (Voorhees and Harman, 2005) therefore emphasize ranking-sensitive measures including Mean Reciprocal Rank (MRR), Normalized Discounted Cumulative Gain (NDCG@k), and Precision@k (Järvelin and Kekäläinen, 2002). Learning-to-rank architectures increasingly optimize directly for such retrieval-oriented objectives in modern recommendation systems (Liu, 2009).

These ranking-oriented evaluation paradigms are especially important in citation prediction because strong pairwise discrimination does not necessarily translate into realistic retrieval performance over large candidate spaces. Citation recommendation systems have progressively moved toward retrieval-oriented pipelines combining semantic search, ranking models, and large candidate pools (Bhagavatula et al., 2018). At large scale, these systems frequently rely on approximate nearest-neighbor retrieval and multi-stage ranking architectures to ensure computational feasibility. Although the present study does not implement neural retrieval or full-corpus approximate nearest-neighbor search, our top- k ranking evaluation over temporally constrained candidate pools provides a rigorous intermediate framework bridging pairwise discrimination and realistic retrieval-oriented evaluation.

3 Methodology

3.1 Mathematical Formalization

We define a citation network as a directed temporally annotated graph $G = (V, E, T)$, where each node $v \in V$ represents a scientific paper with publication timestamp t_v , and each directed edge $(u, v) \in E$ represents a citation from paper u to paper v . The network is temporally acyclic, i.e.,

$$(u, v) \in E \implies t_u > t_v.$$

Temporally Capped Graph. For a citing paper A published at time t_A , we define the temporally capped graph

$$G_{t_A-1} = (V_{<t_A}, E_{<t_A}),$$

where

$$V_{<t_A} = \{v \in V \mid t_v < t_A\}$$

and

$$E_{<t_A} = \{(u, v) \in E \mid t_u < t_A \text{ and } t_v < t_A\}.$$

All features associated with the candidate pair (A, B) are computed exclusively from G_{t_A-1} .

Supervised Learning Objective. Given a candidate pair (A, B) , define

$$y(A, B) = \begin{cases} 1 & \text{if } (A \rightarrow B) \in E, \\ 0 & \text{otherwise.} \end{cases}$$

The supervised learning task is to estimate

$$P(y = 1 \mid \mathbf{x}_{AB}),$$

where $\mathbf{x}_{AB}$ denotes the feature vector associated with the pair (A, B) .

Constrained Candidate Space. For a citing paper A and a true cited paper B , the constrained candidate space is defined as

$$C(A) = \{B' \in V_{<t_A} \mid |t_{B'} - t_B| \leq 3, \text{topic}(B') = \text{topic}(B), (A \rightarrow B') \notin E\}.$$

Ranking Objective. Given a query paper A and a candidate pool

$$\{B\} \cup \{B'_1, \dots, B'_{k-1}\}$$

drawn from $C(A)$, the ranking objective is to assign a score $s(A, \cdot)$ to each candidate such that the true cited paper B is ranked highest. The scoring function $s(A, B)$ is defined as the posterior class-probability estimate returned by the Gradient Boosting classifier.

Ranking quality is evaluated using Mean Reciprocal Rank (MRR) and Normalized Discounted Cumulative Gain (NDCG@ k):

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q(B_q)},$$

$$\text{NDCG@}k = \frac{1}{|Q|} \sum_{q \in Q} \frac{\text{DCG@}k(q)}{\text{IDCG@}k(q)},$$

where Q denotes the set of query papers and $\text{rank}_q(B_q)$ is the rank assigned to the true cited paper B_q for query q .

3.2 A Taxonomy of Temporal Leakage

We define four distinct forms of temporal leakage that commonly affect citation prediction systems.

Future Citation Leakage occurs when the target citation link $(A \rightarrow B)$ is present in the graph used for feature computation. This is eliminated in our framework by removing the target edge prior to feature construction.

Future Topology Leakage occurs when network centrality or overlap measures are computed using nodes and edges entering the network at or after t_A . This allows the model to exploit the future scientific prominence of the cited paper. We eliminate this by restricting all structural computations exclusively to the temporally capped graph G_{t_A-1} .

Metadata Leakage occurs when retrospectively updated metadata are used during prediction. For example, using a journal’s 2024 Impact Factor to predict a citation occurring in 2010 introduces future information unavailable at the time of citation. We mitigate this by using only historically available metadata.

Semantic Leakage occurs when text embeddings derived from pretrained language models implicitly encode citation contexts, particularly if the target papers were included in the pre-training corpus. To reduce this risk, we employ classical TF-IDF cosine similarity rather than transformer-based embeddings. We acknowledge, however, that fitting the TF-IDF vocabulary

on the full corpus introduces minor residual vocabulary leakage through future term-frequency statistics. A fully leakage-free semantic representation would require rolling vocabulary fitting, which we identify as an area for future work.

3.3 Data Collection

We constructed two independent datasets for the discipline of Library and Information Science (LIS), covering the period 1975–2024. These datasets are treated as complementary experiments rather than as a perfectly controlled comparison because their metadata availability differs fundamentally: Dimensions provides a structural-only baseline, whereas OpenAlex enables a semantic+structural framework through abstract availability.

Dimensions Dataset. We queried the Dimensions.ai DSL using the following query:

```
search publications
  where category_for.id = "4610"
  OR title_abstract_only in
    ["knowledge organization", "digital libraries",
     "information literacy", "academic libraries"]
  return publications[id+year+doi+journal+
    references+authors+category_for]
  limit 100000 skip 0
```

After removing records with missing publication years or reference lists, the final dataset contained 259,220 deduplicated articles.

OpenAlex Dataset. We queried the OpenAlex API using five `primary_topic.id` values corresponding to core LIS topics: T10712 (Library and Information Science), T14330 (Digital Libraries), T13166 (Information Retrieval), T13673 (Knowledge Organization), and T10102 (Scholarly Communication). The query was:

```
GET https://api.openalex.org/works?filter=
  primary_topic.id:T10712|T14330|T13166|T13673|T10102,
  type:article,
  publication_year:1975-2024
  &select=id,doi,title,abstract_inverted_index,
    publication_year,referenced_works,authorships,
    primary_topic,cited_by_count
  &per-page=200
```

This yielded 168,901 deduplicated articles with complete abstracts.

3.4 Temporally and Topically Constrained Negatives

Standard link-prediction studies often sample negative edges uniformly at random. In citation networks, this procedure produces trivially separable negatives, such as pairing a contemporary LIS article with an unrelated paper from another discipline or era. We instead employ temporally and topically constrained negatives derived from the candidate space $C(A)$. This forces the model to discriminate between contemporaneously available and topically similar papers,

thereby isolating the actual mechanisms influencing citation choice. This procedure yielded 662,094 balanced pairs for Dimensions and 478,698 balanced pairs for OpenAlex.

3.5 Temporally Aware Feature Engineering

For each candidate pair (A, B) , where A is published in year t_A , all features are computed exclusively from the graph G_{t_A-1} .

Prestige (P_{cited}) is defined as the in-degree of B in G_{t_A-1} , representing the number of citations received by B prior to t_A . **In-degree Velocity** is the proportion of citations to B received during the three years preceding t_A , capturing recent citation acceleration. **Year-Capped PageRank** is the PageRank score of B computed on G_{t_A-1} . In our ablation analysis, PageRank and in-degree velocity provide only modest improvements beyond simple prestige measures; accordingly, we interpret them as baseline enrichment features rather than sophisticated higher-order topological descriptors.

Temporal Gap is defined as the quantity $t_A - t_B$, capturing citation aging effects consistent with the temporal dynamics described by Wang et al. (2013). **Common References** is defined as the quantity $|R_A \cap R_B|$, where R denotes the reference list of a paper, operationalizing bibliographic coupling (Kessler, 1963). **Jaccard References** is the normalized bibliographic-coupling measure $|R_A \cap R_B| / |R_A \cup R_B|$. **Co-citation Overlap** is the number of papers in G_{t_A-1} citing both A and B , operationalizing co-citation (Small, 1973). Because A is newly published at time t_A , this quantity is typically zero unless A previously existed as a preprint.

Semantic Similarity (OpenAlex only) is the TF-IDF cosine similarity between the concatenated title and abstract of A and B . We prefer TF-IDF representations to transformer-based embeddings because they provide greater interpretability, temporal transparency, and reduced contamination risk under leakage-mitigated evaluation. This feature is available only in OpenAlex due to complete abstract availability.

3.6 Model Training and Evaluation

We trained four classifiers: Logistic Regression, Linear SVM, Random Forest, and Gradient Boosting. All models were evaluated using 5-fold stratified cross-validation under a temporal split in which training data consisted of papers published before 2016 and testing data consisted of papers published between 2016 and 2020.

To assess temporal robustness, we additionally evaluated model performance across four rolling temporal windows. For ranking evaluation, we constructed top- k candidate pools with $k \in \{10, 50, 100\}$, and computed MRR, NDCG@ k , and Precision@ k . To quantify uncertainty in the ranking metrics, we estimated 95% bootstrap confidence intervals using 1000 resamples.

4 Results

4.1 Pairwise Classification Performance

We evaluated Logistic Regression, Linear SVM, Random Forest, and Gradient Boosting using 5-fold stratified cross-validation. Table 1 summarizes the predictive performance across the Dimensions and OpenAlex datasets.

Table 1: Cross-validation performance (mean $\pm$ standard deviation across 5 folds).

Model	ROC-AUC	PR-AUC	F1 Score	Brier Score
<i>Dimensions (Structural-Only Baseline)</i>				
Logistic Regression	0.8719 $\pm$ 0.0009	0.8893 $\pm$ 0.0009	0.7757 $\pm$ 0.0010	0.1461 $\pm$ 0.0008
Linear SVM	0.8677 $\pm$ 0.0008	0.8816 $\pm$ 0.0009	0.7701 $\pm$ 0.0010	0.1518 $\pm$ 0.0008
Random Forest	0.8741 $\pm$ 0.0010	0.8858 $\pm$ 0.0009	0.7882 $\pm$ 0.0012	0.1477 $\pm$ 0.0007
Gradient Boosting	0.8949$\pm$0.0006	0.9063$\pm$0.0007	0.8013$\pm$0.0009	0.1307$\pm$0.0004
<i>OpenAlex (Semantic+Structural)</i>				
Logistic Regression	0.9677 $\pm$ 0.0006	0.9710 $\pm$ 0.0006	0.9073 $\pm$ 0.0010	0.0660 $\pm$ 0.0006
Linear SVM	0.9663 $\pm$ 0.0005	0.9698 $\pm$ 0.0005	0.9045 $\pm$ 0.0009	0.0680 $\pm$ 0.0005
Random Forest	0.9693 $\pm$ 0.0006	0.9702 $\pm$ 0.0005	0.9121 $\pm$ 0.0011	0.0646 $\pm$ 0.0007
Gradient Boosting	0.9762$\pm$0.0004	0.9779$\pm$0.0004	0.9219$\pm$0.0009	0.0574$\pm$0.0006

Gradient Boosting consistently outperforms all competing classifiers across both datasets. McNemar’s test confirms that the superiority of Gradient Boosting over Random Forest is statistically significant for both Dimensions ($p < 10^{-25}$) and OpenAlex ($p < 10^{-250}$). The substantially stronger performance of the OpenAlex framework is primarily attributable to semantic similarity features derived from abstracts, a result examined further in the ablation analysis presented in Section 4.4.

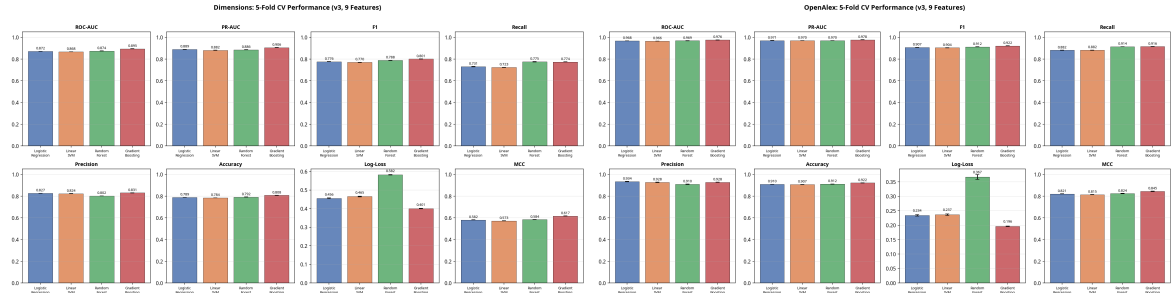


Figure 1: Cross-validation ROC curves for Dimensions (left) and OpenAlex (right). Gradient Boosting achieves the highest ROC-AUC in both datasets.

4.2 Rolling Temporal Evaluation

To evaluate temporal robustness, we assessed the Gradient Boosting classifier across four rolling temporal windows using temporally ordered train–test splits ($t \leq Y_{\text{split}}$ for training and $Y_{\text{split}} + 1$ to Y_{end} for testing).

Table 2: Rolling temporal-window ROC-AUC for Gradient Boosting.

Test Window	Dimensions	OpenAlex
2005–2010	0.8732	0.9662
2010–2015	0.8901	0.9744
2015–2020	0.8948	0.9774
2018–2025	0.8712	0.9788

Performance remains comparatively stable across all temporal windows, indicating that the temporally capped feature engineering generalizes consistently across different publication

periods. The slight decline observed for Dimensions in the 2018–2025 window may reflect shifts in LIS publication behavior or citation density, whereas the gradual increase in OpenAlex performance is plausibly associated with improved abstract completeness in more recent years.

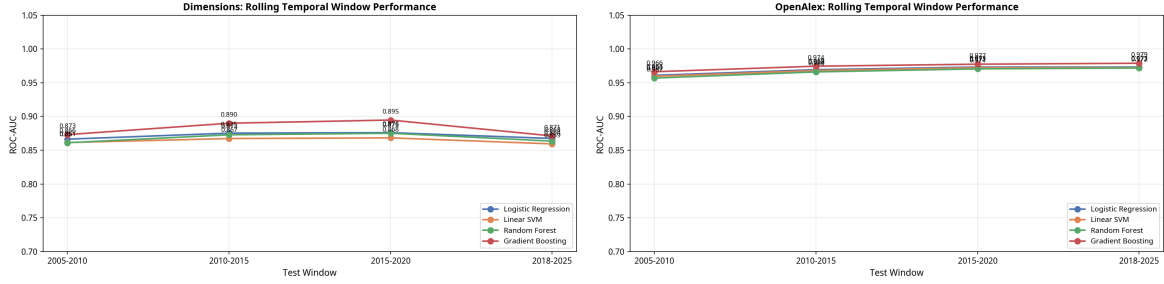


Figure 2: Rolling temporal-window ROC-AUC for Dimensions (left) and OpenAlex (right). Performance remains stable across all four temporal windows.

4.3 Evaluation Taxonomy: From Discriminability to Retrieval Realism

A rigorous evaluation of citation prediction requires distinguishing among multiple levels of task complexity. The first level is **Pairwise Discriminability**, which tests the ability to distinguish a true citation from a single constrained negative example. This isolates feature-level predictive signal but artificially simplifies the candidate space. The second level is **Retrieval Realism (Top- k)**, representing the ability to rank the true cited paper against $k-1$ constrained negatives. This establishes an intermediate evaluation layer between pairwise discrimination and large-scale recommendation.

The third level is **Recommendation Realism**, which measures the ability to rank the true cited paper against the entire historical corpus G_{t_A-1} . This represents the idealized target of citation recommendation systems. We explicitly note that full-corpus retrieval evaluation is computationally infeasible within the present study because of candidate-generation, memory-scaling, and retrieval-latency constraints. Accordingly, we do not claim to approximate industrial-scale recommendation systems. The final level is **Candidate-Space Complexity**, which evaluates the sensitivity of predictive performance to the topical and temporal structure of the negative candidate pool.

While the pairwise classification results in Table 1 evaluate discriminability, realistic citation recommendation operates fundamentally as a ranking problem over competing candidate papers. To evaluate retrieval realism, we therefore constructed top- k ranking tasks in which the true cited paper was ranked against $k-1$ temporally and topically constrained negatives. We additionally report 95% bootstrap confidence intervals (1000 resamples) for the OpenAlex $k=10$ ranking metrics.

Table 3: Top- k ranking evaluation for the 2016–2020 test set.

Dataset & Pool	MRR	NDCG@10	P@1	P@5	Queries
<i>Dimensions (Structural-Only Baseline)</i>					
$k = 10$	0.8088	0.8559	0.7013	0.1926	115,683
$k = 50$	0.5508	0.6137	0.4070	0.1434	3,383
$k = 100$	0.2560	0.3248	0.0723	0.0885	235
<i>OpenAlex (Semantic+Structural)</i>					
$k = 10$	0.9158	0.9369	0.8590	0.1978	96,049
$k = 50$	0.8067	0.8466	0.7059	0.1855	4,312
$k = 100$	0.8213	0.8523	0.7367	0.1878	376
<i>OpenAlex Bootstrap CI ($k = 10$, 1000 resamples)</i>					
MRR	0.9158 [95% CI: 0.9135–0.9181]				
NDCG@10	0.9369 [95% CI: 0.9352–0.9385]				
P@1	0.8590 [95% CI: 0.8551–0.8628]				

The OpenAlex framework maintains strong ranking performance even as the candidate pool expands to $k = 100$, indicating that the high pairwise ROC-AUC translates effectively into retrieval-oriented evaluation over constrained candidate spaces. By contrast, the structural-only Dimensions framework degrades substantially as candidate complexity increases, underscoring the central predictive role of semantic similarity in abstract-rich corpora. We emphasize, however, that these ranking experiments remain constrained by topical and temporal filtering and therefore should not be interpreted as full-corpus recommendation performance.

4.4 Feature Ablation and SHAP Analysis

To identify the principal drivers of predictive performance, we performed leave-one-out feature ablation on the Gradient Boosting models.

Table 4: Feature ablation measured as ROC-AUC decrease after feature removal.

Feature	Dimensions Drop	OpenAlex Drop
Semantic Similarity	N/A	0.0268
Temporal Gap	0.0110	0.0023
Co-citation Overlap	0.0098	0.0029
Prestige (Cited)	0.0061	0.0029
Journal Prestige	0.0036	N/A
Author Productivity	0.0018	N/A
PageRank (Cited)	0.0012	0.0018
In-degree Velocity	0.0005	0.0002

Within the structural-only Dimensions framework, temporal gap, co-citation overlap, and prestige constitute the strongest predictive features. By contrast, PageRank and in-degree velocity contribute only marginal improvements beyond simple prestige measures. We therefore interpret these graph-derived quantities as baseline enrichment features rather than genuinely higher-order topological mechanisms.

In OpenAlex, semantic similarity overwhelmingly dominates the predictive signal. Removing

semantic similarity produces an ROC-AUC decrease of 0.0268, an order of magnitude larger than any other individual feature contribution. A dedicated semantic ablation experiment further confirms this dependence: removing TF-IDF cosine similarity reduces ROC-AUC from 0.976 to 0.949. These results suggest that, in abstract-rich corpora, citation prediction increasingly behaves as a semantic retrieval problem rather than as a purely structural network-prediction task.

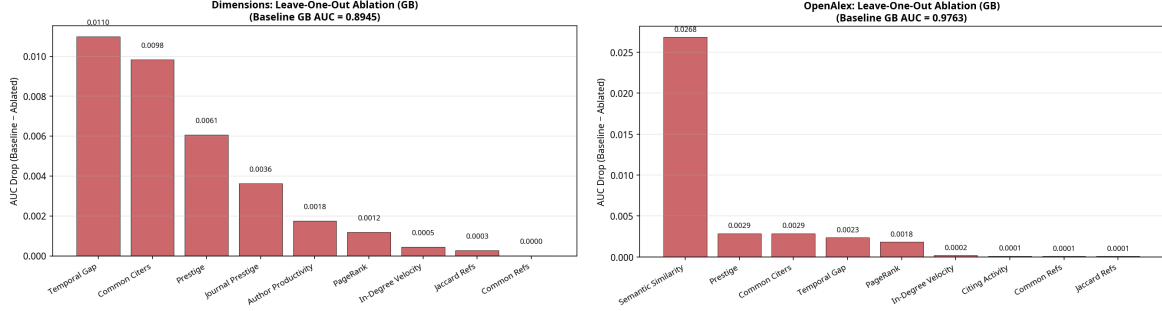


Figure 3: Feature ablation for Dimensions (left) and OpenAlex (right). Semantic similarity dominates predictive performance in OpenAlex, whereas temporal gap and co-citation overlap are the strongest structural features in Dimensions.

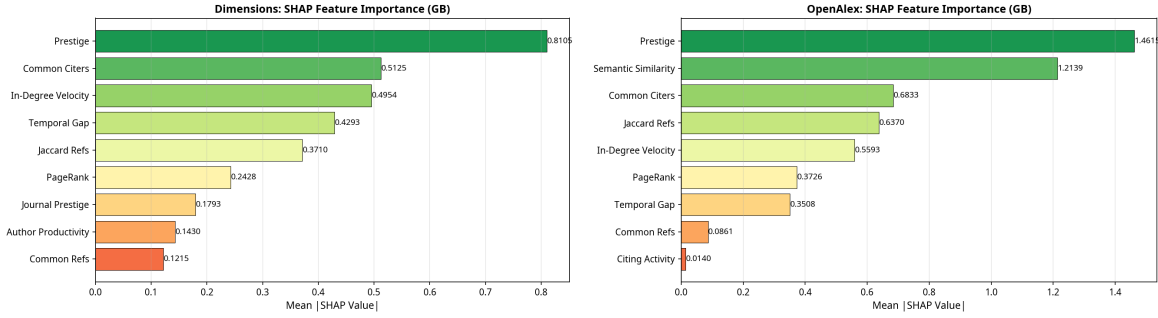


Figure 4: SHAP feature importance for Dimensions (left) and OpenAlex (right). Consistent with the ablation analysis, semantic similarity is the dominant contributor in OpenAlex.

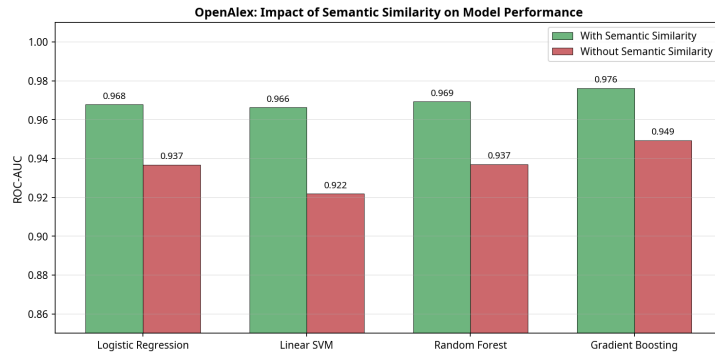


Figure 5: Dedicated semantic ablation for OpenAlex. Removing TF-IDF cosine similarity reduces ROC-AUC from 0.976 to 0.949.

5 Discussion

5.1 Methodological Implications

The primary contribution of the present study is methodological. We have shown that citation link prediction can be evaluated within a strongly leakage-mitigated framework that explicitly respects the temporal structure of citation formation. By combining temporally capped feature engineering with temporally and topically constrained candidate spaces, the framework avoids many of the retrospective artifacts that inflate predictive performance in static citation graphs.

The top- k ranking experiments further demonstrate that the evaluation extends beyond artificially balanced pairwise discrimination into retrieval-oriented settings over constrained candidate pools. Although these experiments remain substantially simpler than full-scale scholarly recommendation systems, the bootstrap confidence intervals indicate that the observed ranking performance is statistically stable under repeated resampling.

The comparison between Dimensions and OpenAlex also reveals an important methodological asymmetry. OpenAlex functions as a semantic+structural framework, whereas Dimensions functions primarily as a structural-only baseline. Because the two datasets differ fundamentally in metadata availability, the comparison should not be interpreted as a perfectly controlled dataset comparison. Rather, the two experiments isolate different sources of predictive signal: the Dimensions framework estimates what can be achieved using structural information alone, whereas OpenAlex demonstrates the substantial additional predictive contribution of semantic similarity once abstracts are available.

5.2 Semantic Dominance and the Identity of Citation Prediction

The ablation experiments reveal that semantic similarity dominates the OpenAlex predictive signal to such an extent that the framework increasingly resembles a semantic retrieval system rather than a purely network-theoretic citation predictor. Removing TF-IDF cosine similarity reduces the OpenAlex ROC-AUC from 0.976 to 0.949, whereas all remaining structural features collectively account for only a comparatively modest fraction of predictive performance. Under leakage-mitigated conditions, citation prediction in abstract-rich corpora therefore appears to function primarily as semantic retrieval augmented by structural baseline enrichment.

Importantly, this finding is not merely tautological. The mean TF-IDF cosine similarity for true citation pairs is 0.0797 (std. 0.0693), compared to 0.0186 (std. 0.0213) for temporally and topically constrained negatives. Only 0.6% of negative pairs exceed a cosine similarity threshold of 0.1, whereas 29.7% of positive pairs do so. The model is therefore not simply identifying near-duplicate documents. Instead, the results suggest that researchers preferentially cite highly specific intellectual precursors occupying narrow semantic neighborhoods rather than sampling broadly across a topic area.

More broadly, these findings imply that citation prediction cannot be understood exclusively as a graph-theoretic problem. In abstract-rich bibliographic infrastructures such as OpenAlex, semantic similarity substantially outweighs higher-order topological structure once temporal leakage is controlled explicitly. The present results therefore support a reframing of citation prediction as a hybrid retrieval problem in which semantics provides the dominant signal and citation topology supplies additional contextual enrichment.

5.3 Interpretive Discussion: Disciplinary Structure and Citation Patterning

The following interpretation should be understood as a theoretical reading of the empirical findings rather than as a directly tested hypothesis. The strong predictive contribution of prestige and structural overlap within the structural-only Dimensions framework indicates that cumulative advantage processes remain highly influential within LIS citation behavior. Papers are cited not solely because of their semantic content, but also because they occupy structurally visible and intellectually proximate positions within the evolving citation network.

This high degree of predictability suggests a broader interpretation concerning disciplinary organization and knowledge diffusion. Citation behavior within LIS appears strongly patterned by existing intellectual structure, indicating that researchers search for relevant work within relatively constrained semantic and structural neighborhoods. Such a pattern is consistent with sociological theories emphasizing invisible colleges, disciplinary cohesion, and cumulative scientific development.

At the same time, an important caveat is necessary. Paradigmatic closure, bounded disciplinary search, and the cognitive structures discussed by Kuhn (1962) and Crane (1972) are not directly operationalized within the present feature space. Our variables measure structural and semantic proximity, but they do not directly capture the cognitive or institutional mechanisms through which disciplinary paradigms constrain scholarly attention. The sociological interpretation advanced here should therefore be understood as an interpretive overlay rather than as a direct analytical inference from the model itself.

5.4 Limitations and Future Work

Several limitations should be acknowledged explicitly. First, the TF-IDF representations were fitted on the full corpus vocabulary, introducing minor residual vocabulary leakage through future term-frequency statistics. A fully leakage-free semantic representation would require rolling vocabulary construction across temporal windows.

Second, although the top- k ranking experiments approximate retrieval-oriented evaluation over constrained candidate pools, realistic scholarly recommendation systems operate over substantially larger candidate spaces involving millions of documents. Full historical-corpus evaluation introduces complex candidate-generation, indexing, and memory-scaling challenges that remain outside the scope of the present study.

Third, the two bibliographic infrastructures differ substantially in metadata availability, making perfectly controlled comparisons impossible. Future work should attempt to harmonize feature spaces across datasets in order to isolate more precisely the relative contributions of semantic and structural information.

Finally, the present experiments are confined primarily to Library and Information Science. Although we executed preliminary robustness checks using Physics and Computer Science corpora, citation cultures vary substantially across disciplines, and LIS may possess structural characteristics that make citation behavior comparatively predictable. Broader cross-disciplinary evaluation therefore remains an important direction for future work.

6 Conclusion

This study introduced a strongly leakage-mitigated methodological framework for citation link prediction grounded explicitly in the temporal structure of scholarly communication. By constructing temporally capped citation graphs, restricting feature computation to historically available information, and evaluating against temporally and topically constrained candidate spaces, we demonstrated that citation prediction can be evaluated under substantially more rigorous conditions than those typically employed in static graph-based benchmarks. In doing so, we formalized multiple forms of temporal leakage—including future citation leakage, future topology leakage, metadata leakage, and semantic leakage—and operationalized concrete mitigation strategies within a unified evaluation framework.

Beyond pairwise classification, we further demonstrated that citation prediction should be understood fundamentally as a retrieval-oriented problem. The incorporation of top- k ranking evaluation provides a principled intermediate layer between binary discrimination and large-scale recommendation systems operating over massive bibliographic corpora. Although our experiments remain constrained to temporally and topically filtered candidate pools rather than unrestricted historical retrieval, the ranking results show that leakage-mitigated citation prediction retains substantial predictive power under increasingly demanding candidate-space conditions.

The comparison between the Dimensions and OpenAlex frameworks also revealed an important methodological and empirical asymmetry. The structural-only Dimensions experiments demonstrate that prestige, temporal proximity, and structural overlap remain meaningful predictors of citation behavior under temporally constrained evaluation. At the same time, the OpenAlex experiments show that once abstract-level semantic information becomes available, semantic similarity overwhelmingly dominates predictive performance. Under these conditions, citation prediction increasingly behaves less like a purely graph-theoretic link-prediction problem and more like a semantically guided retrieval process augmented by structural contextualization.

More broadly, the present findings suggest that scholarly citation behavior is highly organized both semantically and structurally. Even after restricting the evaluation to historically available information and challenging constrained negatives, citation formation remains strongly predictable. This does not imply deterministic citation behavior, nor does it establish direct evidence for paradigmatic closure or bounded disciplinary cognition. Nevertheless, it indicates that scientific attention is concentrated within relatively stable semantic and structural neighborhoods that can be modeled with considerable accuracy under temporally rigorous conditions.

Taken together, the present study contributes not merely a predictive model, but a methodological framework for evaluating citation prediction under explicit temporal constraints. We argue that future bibliometric machine-learning systems should be assessed not only by predictive accuracy, but also by the historical validity of the information available at prediction time. In this sense, temporally rigorous evaluation is not simply a technical refinement of citation prediction, but a necessary condition for its scientific interpretability.

Data and Code Availability

All data processing pipelines, feature engineering scripts, model training code, and result files reported in this study are publicly available at:

The repository includes the scripts required to reproduce all cross-validation, rolling temporal window, ablation, and ranking experiments reported in this paper, as well as the code generating all figures.

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